

Problem 26.11

If X is the present value of a perpetuity of 1 per year with the first payment at the end of the 2nd year and $20X$ is the present value of a series of annual payments $1, 2, 3, \dots$ with the first payment at the end of the 3rd year, find d .

Solution

A perpetuity-immediate of 1 per year has present value

$$a_{\infty|} = \frac{1}{i}.$$

Since its first payment is deferred until the end of the 2nd year,

$$X = va_{\infty|} = \frac{v}{i}.$$

The increasing perpetuity $1, 2, 3, \dots$ normally begins at the end of the first year and has present value

$$(Ia)_{\infty|} = \frac{1}{id}.$$

Here its first payment is at the end of the 3rd year, so it is deferred two years. Therefore,

$$20X = v^2(Ia)_{\infty|} = \frac{v^2}{id}.$$

Substitute $X = v/i$:

$$20\left(\frac{v}{i}\right) = \frac{v^2}{id}.$$

Canceling v/i gives

$$20 = \frac{v}{d},$$

and hence

$$20d = v.$$

Since $v = 1 - d$,

$$20d = 1 - d.$$

Thus,

$$21d = 1,$$

so

$$\boxed{d = \frac{1}{21}}.$$

Numerically,

$$d = 0.04761905.$$